Rank-27 Vivify Arena-Order Representation Repair

Tactix Development Project

25 August 2026

Executive summary

This increment promotes a general representation repair in Vivify without changing Vivify's semantics. Tactix allocates a fresh variable-width arena record when Vivify strengthens a clause, then restores the surviving database to inherited logical arena order. The old implementation used `std::stable_sort`; every comparison recovered both clauses' logical keys. Kernel already owned an exact cached-key stable LSD radix pass for the same order. Vivify now delegates to that primitive.

Candidate order, rule and scheduling order, exact logical ticks and costs, proof effects, learning, pruning, retained state, and N1–N4 are unchanged. The production source diff has SHA-256 `6 5e9489b66344023c245a1450b12e9d76fa178e7926cf68cf5ac0f02ce442824`; the validated Release binary has SHA-256 `067f7f3a0566cce6a52c90d78228b51cc7444adadb464402d5339f07ed5070bd`. No benchmark identity, rank, deadline, per-instance arm, or tuned constant is present in the solver.

The exact conflict-9450 stage lowers Vivify 2.106 to 1.865 seconds. The complete frozen B28 plus mechanism-relevant C5 gate passes 66/66 independently checked arms and lowers aggregate Vivify 18.07%, with candidate/shipped PAR-2 ratio 0.978814. A checked Rank-27 proof witness and Claude-owned exact-source sanitizer/native-LRAT fuzz both pass with zero residue.

An apparent Rank-27 regression from the accepted approximately 399 seconds to over 500 seconds required a separate causal audit before promotion. The unchanged accepted predecessor also reproduced approximately 580 seconds. The final two-node crossover then measured both binaries once on each node: node-local candidate deltas had opposite signs, while the balanced candidate average was 2.198 seconds faster, Vivify was 11.52% faster, and inprocessing was 7.744 seconds faster. Thus the current 399-to-580-second absolute drift is environmental, not caused by the Vivify repair. The repair is promoted; Rank 27 and broader convergence remain open.

1 Repair and exact common-state evidence

Clause order is semantic scheduling state in Tactix. New large clauses use a monotone proof identity as their logical order key, while restatements and packed replacements inherit the source key. Vivify's collection hook must therefore restore the exact stable sequence ordered by logical key and then by the existing Ref. The repair exposes Kernel's existing cached-key radix primitive and calls it from `VivifyArenaOrder`; physical arena addresses remain neither proof authority nor strategy input.

Exact paired stage `1646181` keeps every normalized non-time line identical and changes Vivify 2.106 to 1.865 seconds (0.88557x), inprocessing 16.019 to 15.775 seconds, and wall time 16.51 to 16.25 seconds. Peak RSS is unchanged. Event job `1646183` emits all 64 untruncated rows with the frozen byte-identical SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`. This directly freezes rule identity, candidate and scheduling order, exact tick/cost fields, boundary state, learning, pruning, and resulting state.

The full-width frozen B28+C5 campaign `1646184/1646250` validates 17 SAT models and 16 UNSAT proofs on each arm. All 33 paired normalized non-time images are exact; invalidity, solve loss, cleanup failure, and certificate residue are zero. Aggregate Vivify falls 198.247 to 162.431 seconds (0.81933x), and candidate PAR-2 is 0.978814x shipped. Aggregate summary SHA-256 is `18a60d2f61ef9281dabe6e6fb9aa976f34b5bb7970e9c3f8f801e00d79e5eb6c`.

Proof-producing Rank-27 job `1646346` returns UNSAT on the exact accepted trajectory. Pinned `drat-trim` verifies the 1,514,395,111-byte proof, SHA-256 `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`, before all proof, decompressed input, and checker scratch are deleted. Claude Code's authoritative terminal report for exact-source job `1646352` is PASS: corpus 27/27 and random 10,000/10,000, with 4,966 checked SAT models, 3,451 checked UNSAT proofs, 1,583 declared bounded nonterminations, direct reach in 5,865 Vivify and 4,306 changed-Vivify cases, zero sanitizer/crash/ invariant failure, and zero residue.

2 Resolving the apparent whole-run regression

Sequential Rank-27 pair `1646251` lowered Vivify 64.399 to 58.191 seconds and inprocessing 192.415 to 186.927 seconds, but total time moved 515.505 to 520.897 seconds. That pair could not price the change: its unchanged control was already roughly 116 seconds slower than the accepted 399.152-second run. A separate candidate-only run `1647203` then took 583.796 seconds, so the repair was reopened under the user-directed 500-second criterion.

The identical standalone wrapper with the immutable unchanged predecessor, job `1647229`, took 580.077 seconds. Both runs used their CPUs fully and all non-time lines were byte-identical. Candidate versus predecessor was only +3.719 seconds, while candidate Vivify improved 70.742 to 60.910 seconds and inprocessing improved 204.559 to 194.157 seconds. This disproved Vivify as the cause of the absolute jump, but a clean total comparison was still needed. Exclusive-node submission was rejected by the partition before allocation, so no solver or artifact was created.

Final campaign `benchmarks/runs/20260825-0840-tierd-rank27-vivify-radix-cross-node-control/`, job `1647252`, reserves one two-CPU/16 GiB task on each available node. Phase 1 runs predecessor on rank 0 and candidate on rank 1; phase 2 swaps the binaries without changing rank-to-node placement. No two Tactix processes share a node. All four runs return exact-state UNSAT at 99% CPU with zero residue.

Placement	Predecessor	Candidate	Candidate delta
node017 total, s	583.044	556.510	−26.534
node018 total, s	450.091	472.229	+22.138
balanced total, s	516.5675	514.3695	−2.198
balanced Vivify, s	64.5305	57.0970	−7.4335
balanced inprocessing, s	191.2535	183.5095	−7.744

The opposing node-local signs show that node/phase conditions dominate any single total comparison. Under the additive crossover estimator, node and phase biases cancel: candidate total is 0.995745x predecessor, Vivify is 0.884807x, and inprocessing improves 7.744 seconds. The remaining outside- inprocessing average moves by 5.546 seconds, smaller than the environmental crossovers and not accompanied by any state change. The approximately 399-second historical result is currently reproducible by neither binary and cannot be used to reject this exact-state repair.

3 Disposition and next work

The final BVE-free Release build reproduces the validated binary pin. The complete suite passes 73/74: every executable, solver, proof/model, trace, campaign-cleanup, proof-rule/strategy-boundary, and solve-reset test passes; only the already-recorded opt-in documentation-size policy fails. Frozen reference solvers were not rerun and all campaign certificate residue is zero.

The Vivify radix arena-order repair is therefore promoted. It does not claim that Vivify convergence is globally complete; it closes one portable common- state representation owner. Rank 27 remains above the 300-second milestone, and reaching that milestone will not stop justified convergence. The paused residual-BVE flat-slice prototype is preserved separately and is not part of this source. After this checkpoint lands cleanly, resume that staged BVE comparison, then continue broad search quality/throughput localization. The accepted Sweep, directed bridge, BVE antecedent-admission, Congruence, and prior Vivify semantic closures remain closed absent contradictory exact-state evidence.